Scalability & Future Plan

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Team ID	AIML-HDI-2026
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Scalability & Future Plan

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Secondary-feature data leakage	The model's 10 secondary features include HDI Rank and historical HDI values (1990–1998), meaning part of the model's accuracy comes from HDI predicting HDI rather than independent indicators.	High
2	Single-Node / Free-Tier Deployment	Render's free tier causes “cold starts” after ~15 minutes of inactivity and limits concurrent user requests.	High
3	Static Training Data	The model is trained on a 2021 snapshot; predictions may gradually lose real-world accuracy as global socio-economic conditions shift.	Medium
4	High-Tier Classification Accuracy	The “High” development tier has the lowest precision (0.63) of the four tiers, due to boundary confusion with “Very High.”	Medium
5	Lack of User Accounts	Users cannot save their prediction history, compare past inputs, or log back in to view previous reports.	Low

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Hosted on a single free-tier Render instance.	Upgrade to a paid cloud tier (e.g., Render Pro or AWS EC2) and enable auto-scaling to handle traffic spikes.
2	Data Storage	Stateless architecture; no user data or prediction logs are saved	Integrate a relational database (e.g., PostgreSQL) to store prediction history

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
		persistently.	and support future dataset updates.
3	Model Freshness	Model trained once on a static 2021 dataset snapshot.	Automate periodic retraining as new UNDP HDI releases become available.
4	Performance	Synchronous request handling via Flask routing.	Introduce async task queuing (e.g., Celery + Redis) for bulk predictions or future PDF report generation.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Retrain the secondary branch on genuinely independent indicators (health, environment, inequality) instead of historical HDI/rank.	1–2 Months	Removes the data-leakage concern and produces a more defensible, generalizable model.
Phase 3	Add explainability (e.g., feature attribution) to the results dashboard.	2–3 Months	Lets users see which input most influenced a given prediction, building trust in the tool.
Phase 4	Move to a paid, always-warm hosting tier.	As needed / on demo day	Eliminates cold-start delay for live demonstrations and real users.
Phase 5	Automated PDF report generation from a prediction.	3–4 Months	Allows users to instantly download a formatted HDI analysis report based on their inputs.